Joint Distributions

Bivariate distribution

Definition

X, Y: discrete random variables defined on the same sample space.

A, B: sets of possible values of X and Y.

$$p(x,y) = P(X = x, Y = y) = P(A \cap B)$$

$$\sum_{x \in A} \sum_{y \in B} p(x,y) = 1$$

Probability mass function of X

$$p(x) = P(X = x) = P(x = x, y \in B) = \sum_{y \in B} p(x,y)$$

Theorem

p(x, y): Joint probability mass function of discrete R.V. X and Y.

h(X, Y): A function of X and Y

$$E[h(X,Y)] = \sum_{x \in A} \sum_{y \in B} h(x,y)p(x,y)$$

$$E(X+Y) = E(X) + E(Y)$$

Proof

$$\begin{aligned} E[X+Y] &= \sum_{x \in A} \sum_{y \in B} (x+y)p(x,y) \\ &= \sum_{x \in A} \sum_{y \in B} xp(x,y) + \sum_{x \in A} \sum_{y \in B} yp(x,y) \\ &= \sum_{x \in A} x(\sum_{y \in B} p(x,y)) + \sum_{y \in B} y(\sum_{x \in A} p(x,y)) \\ &= \sum_{x \in A} xp_X(x) + \sum_{y \in B} yp_Y(y) = E[X] + E[Y] \end{aligned}$$

Definition

Two r. v.'s X and Y, defined on the same sample space, have a continuous joint distribution if there exists a nonnegative function of 2 variables, f(x,y) on $R \times R$ such that for some region S in the xy-plane

$$P((X,Y) \in S) = \iint_S f(x,y)dxdy$$

$f(x,y)$: is called the joint (probability) density function of X and Y.

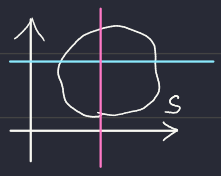
Definition

單點機率為 0：$P(X=a,Y=b)=\int_b^b\left(\int_a^a f(x,y)dx\right)dy=0$

總和機率為 1：$A=(-\infty,+\infty),B=(-\infty,+\infty)\implies\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}f(x,y)dxdy=1$

Definition

Let X and Y have joint density function f(x,y)



$f_Y(y)=\int_{-\infty}^{\infty}f(x,y)dx$

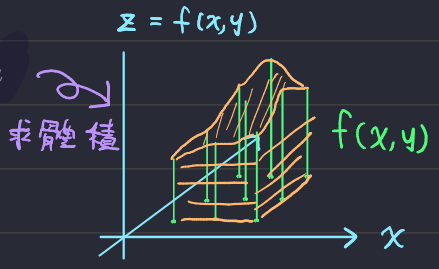
$f_X(x)=\int_{-\infty}^{\infty}f(x,y)dy$

are called, the marginal (probability) density functions of X and Y.

Definition

Joint (cumulative) distribution function

$F(X,Y)=F(t,u)=P(X\le t,Y\le u)=\int_{-\infty}^{y}\int_{-\infty}^{x}f(u,t)dydu$



The marginal distribution function

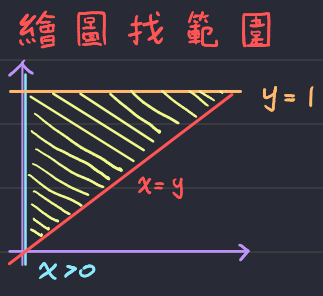
$F_X(t)=P(X\le t)=P(X\le t,Y\le\infty)=F(t,\infty)$

$F_Y(t)=P(Y\le t)=P(X\le\infty,Y\le t)=F(\infty,t)$

Ex 8.4 The joint density function of random variables X and Y is given by

$$f(x,y)=\begin{cases}\lambda xy^2 & 0\le x\le y\le 1\\ 0 & \text{otherwise.}\end{cases}$$

(a) Determine the value of lambda



$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}f(x,y)\,dx\,dy$

$=\int_0^1\int_0^y\lambda xy^2\,dx\,dy$

$=\int_0^1\lambda\left(\frac{1}{2}x^2y^2\right)\Big|_0^y dy$

$=\lambda\int_0^1\frac{1}{2}y^4\,dy$

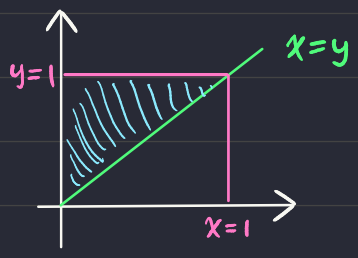
$=\lambda\left(\frac{1}{10}y^5\right)\Big|_0^1$

$=\frac{1}{10}\lambda$

Ans: $\lambda=10$ #

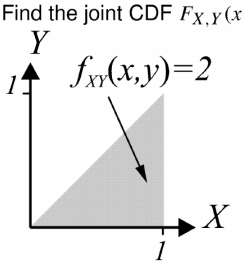
(b) Find the marginal density function of X and Y.　　作答注意範圍

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$$
$$= \int_{x}^{1} f(x,y)\,dy$$
$$= 10\left(\frac{1}{3}xy^3\right)\Big|_x^1$$
$$= \frac{10}{3}(x - x^4)\ \#\quad \text{if } 0 \le x \le 1$$



$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$$
$$= \int_{0}^{y} 10xy^2\,dx$$
$$= 5x^2y^2\Big|_0^y$$
$$= 5y^4\ \#\quad \text{if } 0 \le y \le 1$$

Find the joint CDF $F_{X,Y}(x, y)$ when X and Y have joint PDF



$$f_{X,Y}(x, y) = \begin{cases} 2 & 0 \le y \le x \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

$$F(t,u) = P(X \le t,\ Y \le u)$$
$$= \int_{-\infty}^{Y}\int_{-\infty}^{X} f(x,y)\,dt\,du$$
$$= \int_{0}^{Y}\int_{u}^{X} 2\,dt\,du$$
$$= \int_{0}^{Y} (2t)\Big|_u\,du$$
$$= \int_{0}^{Y} 2X - 2u\,du$$
$$= (2Xu - u^2)\Big|_0^Y$$
$$= 2XY - Y^2$$

後面還有一把範圍判定



Ex 8.5

For $\lambda > 0$, let

$$F(x,y) = \begin{cases} 1 - \lambda e^{-\lambda(x+y)} & \text{if } x > 0,\ y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

Determine if F is the joint distribution function of 2 random variables X and Y.

$\frac{\partial^2}{\partial x \partial y} F(x,y)$

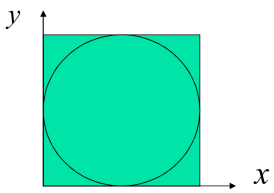
$= \frac{\partial}{\partial x} -\lambda e^{-\lambda(x+y)} \cdot (-\lambda)$

$= \frac{\partial}{\partial x} \lambda^2 e^{-\lambda(x+y)}$

$= -\lambda^3 e^{-\lambda(x+y)}$ if $x>0$, $y>0$

↳ 負的，所以不行 ∴ #

Ex 8.6 **A circle of radius 1 is inscribed in a square with sides of length 2. A point is selected at random from the square. What is the probability that the point is inside the circle?**



$\frac{\pi}{4}$ #

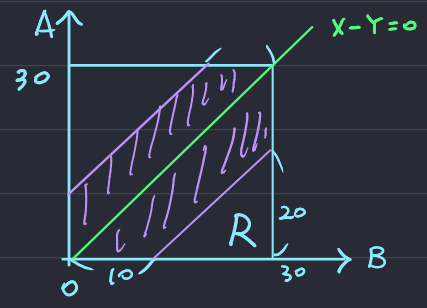
Definition

Let S be a subset of the plane with area A(S)

A point is said to be randomly selected from S if for any subset R of S with area A(R), the probability that R contains the point is A(R)/A(S).

Ex 8.7 **A man invites his fiancee to a fine hotel for a Sunday brunch. They decide to meet in the lobby of the hotel between 11:30 A.M. and 12 noon. If they arrive at random times during this period, what is the probability that they will meet within 10 minutes?**

面積法：



R: possible meet time

S: $|X-Y| \leq 10$

A: $30 \times 30 - 2 \times (20 \times 20 \times \frac{1}{2})$

$= 500$

ANS: $\frac{500}{900}$

Theorem

$Z = h(x,y)$: The actual value of the function

$f(x,y)$: The p.d.f. of continuous R.V. X and Y

$$E(Z) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y) f(x,y)\, dxdy$$

$$E(X+Y) = E(X) + E(Y)$$

Proof

$$\begin{aligned} E(X+Y) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x+y) f(x,y) dxdy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y) dxdy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x,y) dxdy \\ &= E(X) + E(Y). \end{aligned}$$

Theorem

$$E(Z) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y)f(x,y)dxdy$$

$Z = h(x,y)$

$f(x,y)$: The probability density function

Corollary

For continuous r. v.'s X and Y, $E(X+Y) = E(X) + E(Y)$

$$E(X+Y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)f(x,y)dxdy \qquad \text{marginal density function}$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xf(x,y)dxdy + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} yf(x,y)dxdy$$
$$= E(X) + E(Y).$$

Ex 8.9 Let X and Y have joint density function

$$f(x,y) = \begin{cases} \frac{3}{2}(x^2+y^2) & \text{if } 0 < x < 1, 0 < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find $E(X^2+Y^2)$

$h(X,Y) = X^2+Y^2$

$E(X^2+Y^2) = E(h(X,Y)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(x,y)\cdot f(x,y)\,dx\,dy$

$= \int_0^1\int_0^1 (X^2+Y^2)\cdot\frac{3}{2}(X^2+Y^2)\,dx\,dy$

$= \frac{3}{2}\int_0^1\int_0^1 X^4+2X^2Y^2+Y^4\,dx\,dy$

$= \frac{3}{2}\int_0^1 \left(\frac{1}{5}X^5+\frac{2}{3}X^3Y^2+Y^4X\right)\Big|_0^1$

$= \frac{3}{2}\int_0^1 \left(\frac{1}{5}+\frac{2}{3}Y^2+Y^4\right)dy$

$= \frac{3}{2}\left(\frac{1}{5}Y+\frac{2}{9}Y^3+\frac{1}{5}Y^4\right) = \frac{3}{2}\cdot\frac{28}{45} = \frac{14}{15}$ #

8.2 Independent random variables

Definition

A, B are called independent if $P(\in A, Y \in B) = P(X \in A)P(Y \in B)$

If A, B are independent, then :

Probability	$P(X \le a, Y \le b) = P(X \le a)P(Y \le b)$	
Joint distrubution function	$F(t,u) = P(x < t)P(y < u) = F_x(t)F_y(u)$	
Joint PMF	$p(x,y) = p_x(x)p_y(y)$	離散
Joint density function	$f(x,y) = f_x(x)f_y(y)$	連續

Theorem 8.5

X, Y: two independent R.V.

g, h: two R → R function

→ g(x) and h(Y) are also independent

Theorem 8.6

X, Y: two independent R.V

g, h: two R → R function

$E[g(X)h(Y)] = E[g(x)]E[h(Y)]$

本理論不是雙向的，如果

$E[g(X)h(Y)] = E[g(x)]E[h(Y)]$

成立，不代表他們一定獨立

X: a uniform distribution in [-1, 1]

Y = X^2

E(X) = 0 E(X)*E(Y) = 0

E(XY) = E(X * X^2) = E(X^3) = 0

Proof $E(Z) = E[g(x)h(y)]$

$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} [g(x)h(y)] \cdot f(x,y)\,dx\,dy$

$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} [g(x)h(y)] \cdot [f_x(x) \cdot f_y(y)]\,dx\,dy$

$= \int_{-\infty}^{\infty} h(y) f_y(y)\,dy \cdot \int_{-\infty}^{\infty} g(x) f_x(x)\,dx$

$= E[h(y)] \cdot E[g(x)]$

Ex 8.12 **Store A and B, which belong to the same owner, are located in 2 different towns. If the density function of the weekly profit of each store, in thousands of dollars, is given by**

$$f(x)=\begin{cases} x/4 & \text{if } 1<x<3 \\ 0 & \text{otherwise,} \end{cases}$$

and the profit of one store is independent of the other, what is the probability that next week one store makes at least \$500 more than the other store?

$P(X > Y+\frac{1}{2}) + P(Y > X+\frac{1}{2}) = 2P(X > Y+\frac{1}{2})$

$f_X(x) = \begin{cases} \frac{1}{4}x & \text{if } 1<x<3 \\ 0 & \text{otherwise} \end{cases}$ $f_Y(y) = \begin{cases} \frac{1}{4} & \text{if } 1<x<3 \\ 0 & \text{otherwise} \end{cases}$

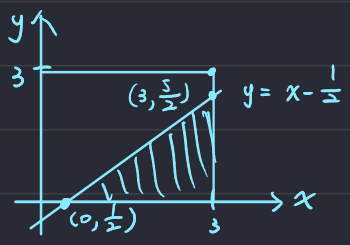
$f(x,y) = \begin{cases} \frac{1}{16}xy & \text{if } 1<x<3,\ 1<y<3 \\ 0 & \text{otherwise} \end{cases}$

$\int_{\frac{1}{2}}^{3} \int_{0}^{x-\frac{1}{2}} f(x,y)\,dy\,dx$

$= \int_{\frac{1}{2}}^{3} \int_{0}^{x-\frac{1}{2}} \frac{1}{16}xy\,dy\,dx$

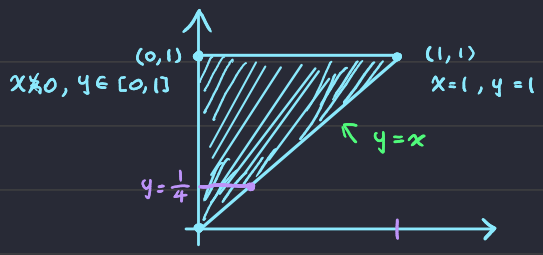
$= \cdots$ (省略千多)

$= \frac{549}{1024}$



Ex 8.15 **Prove that 2 random variables X and Y with the following joint density function are not independent.**

$$f(x,y)=\begin{cases} 8xy & 0 \le x \le y \le 1 \\ 0 & \text{otherwise.} \end{cases}$$



$f_X(x) = \int_x^1 8xy\,dy$

$= 4xy^2 \Big|_x^1 = 4x - 4x^3$

$f_Y(y) = \int_0^y 8xy\,dx$

$= 4x^2y \Big|_0^y = 4y^3$

找 x 範圍

1. 找隨意 y 值
2. 看 x 的 Range
3. 0 ~ y

$f(x,y) \neq f_X(x) \cdot f_Y(x)$ #

8.3 Conditional distributions

Definition

The conditional probability mass function of X given that Y=y

$$P_{X|Y}(X|y) = P_X(X,Y) = P(X|Y) = P(X=x \mid Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)} = \frac{P(x,y)}{P_y(y)}$$

$$\sum_{x\in A} = \sum_{x\in A} \frac{p(x,y)}{y_y(y)} = 1$$

$$F_{X|Y}(x|y) = P(X < x|Y = y) = \sum_{t\le x} P(X = t, Y = y) = \sum_{t\le x} p_{X|Y}(t|y)$$

微分

$$f_{X|Y}(x,y) = \frac{f(x,y)}{f_Y(y)}$$

If X and Y are two independent variable

$$p_{X|Y}(x|y) = p(x)$$

$$f_{X|Y}(x|y) = f_X(x)$$

Ex 8.21 **Let X and Y be continuous random variables with joint density function**

$$f(x,y)=\begin{cases}\frac{3}{2}(x^2+y^2) & \text{if } 0<x<1,\ 0<y<1\\ 0 & \text{otherwise.}\end{cases}$$

Find $f_{X|Y}(x\,|\,y)$.

$f(x|y)=\frac{f(x,y)}{f(y)}$

$f(y)=\int_0^1 \frac{3}{2}(x^2+y^2)\,dx$

$=\frac{3}{2}y^2+(\frac{1}{2}x^3)\Big|_0^1$

$=\frac{3}{2}y^2+\frac{1}{2}$

$$f_{X|Y}(x\,|\,y)=\frac{3/2(x^2+y^2)}{(3/2)y^2+1/2}=\frac{3(x^2+y^2)}{3y^2+1}$$

for $0<x<1$ and $0<y<1$.

Everywhere else, $f_{X|Y}(x\,|\,y)=0$.

注意範圍

Ex 8.22 **First, a point Y is selected at random from the interval (0,1).Then another point X is chosen from the interval (0,Y). Find the density function of X.**

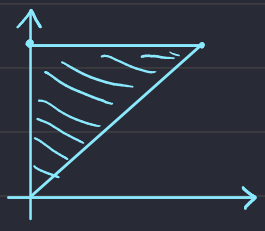
$f(Y)=\begin{cases}1 & \text{if } 0\le y\le 1\\ 0 & \text{otherwise}\end{cases}$

$f(X|Y)=\begin{cases}\frac{1}{y} & \text{if } 0\le x\le y,\ 0\le y\le 1\\ 0 & \text{otherwise}\end{cases}$

$f(x|Y)=\frac{f(x,y)}{f(Y)}$

$f(x,y)=f(x|Y)\cdot f(Y)$

$=\frac{1}{y}$ if $0\le x\le y\le 1$



$f(x)=\int_x^1 \frac{1}{y}\,dy$

$=-\ln(x)$ if $0\le x\le 1$, elsewhere 0 #

Ex 8.18 **Calculate the expected number of aces in a randomly selected poker hand that is found to have exactly 2 jacks.**

5 cards

X : The number of ace

Y : The number of jack

$$E(X \mid Y=2) = \sum_{x=0}^{3} x\, p(x \mid Y=2)$$

3 → J J 2 1 7

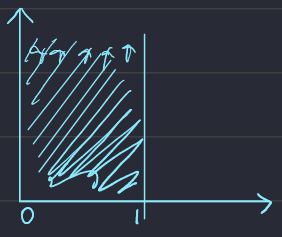
$$= \sum_{x=0}^{3} x \cdot \frac{C_x^4 \cdot C_2^4 \cdot C_{5-x-2}^{44} / C_5^{52}}{C_2^4 \cdot C_3^{48} / C_5^{52}}$$

$$\approx 0.25 \ \#$$

Ex 8.24 **Let X and Y be continuous random variables with joint density function**

$$f(x,y) = \begin{cases} e^{-y} & \text{if } y > 0,\ 0 < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

Find $E(X \mid Y = 2)$.



$$E(X \mid Y=2) = \int_{-\infty}^{\infty} x \cdot f(x,2)\, dx$$

$$f(x,2) = \frac{f(x,2)}{f_y(2)}$$

$$f_y(2) = \int_{-\infty}^{\infty} f(x,2)\, dx = \int_0^1 e^{-2}\, dx = e^{-2}$$